

DIGITAL FORENSICS

FINAL INVESTIGATION REPORT

Formatted USB Data Recovery Using Magnet AXIOM

Case Reference No.	DFIR-2026-RMF-011
Investigator	Julian Derry
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1. Overview

This examination involved the forensic acquisition and analysis of a formatted USB flash drive using Magnet AXIOM Process and Magnet AXIOM Examine. The objective was to determine whether previously stored data remained recoverable after formatting and to document the acquisition, processing, examination, and verification workflow in a forensically sound manner.

The investigation demonstrated that residual data artifacts were still present on the formatted media and could be reconstructed through forensic processing. The case highlights an important digital forensic principle: formatting a storage device does not necessarily result in permanent data destruction.

2. Case Initialization

A new case (Case 011) was created in Magnet AXIOM Process. The formatted USB flash drive was connected and identified as a removable storage device. Device details were reviewed to confirm the correct evidence source before analysis began. A designated folder, named according to the case type, was created to store case files and acquired evidence for proper organization and easy identification.

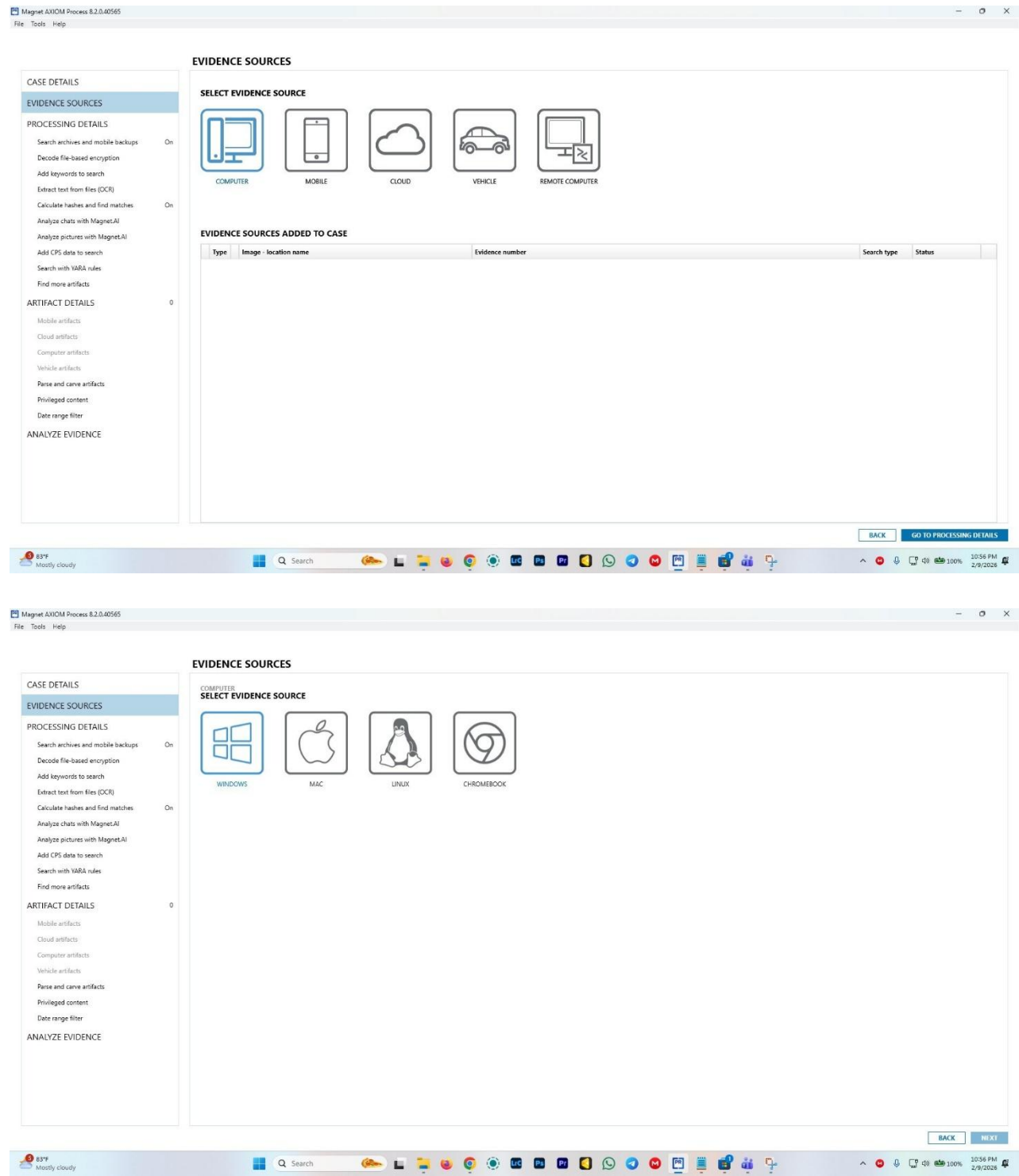
The screenshot displays the Magnet AXIOM Process 8.2.0.40565 application window. The interface is divided into a left sidebar and a main content area. The sidebar contains a 'CASE DETAILS' tab and a list of evidence sources under 'EVIDENCE SOURCES'. The main content area is titled 'CASE DETAILS' and contains several sections: 'CASE INFORMATION' with fields for Case number (001) and Case type (Intrusion / Incident response); 'LOCATION FOR CASE FILES' with fields for Folder name (AXIOM - Feb 09 2026 202754) and File path (C:\Users\MK52\Desktop\Axiom Forensics\Intrusion - Incident response); 'LOCATION FOR ACQUIRED EVIDENCE' with fields for Folder name (AXIOM - Feb 09 2026 202754) and File path (C:\Users\MK52\Desktop\Axiom Forensics\Intrusion - Incident response); 'SCAN INFORMATION' with fields for Scanned by (Julian Derry) and Description (Recovering deleted files from a flashdrive); and 'REPORT OPTIONS' with a field for Cover logo (C:\Users\MK52\Pictures\logo 2.png). A 'GO TO EVIDENCE SOURCES' button is located at the bottom right of the main content area.

3. Evidence Source Selection

The USB flash drive was selected as the sole evidence source by choosing Computer from the list of available evidence sources.

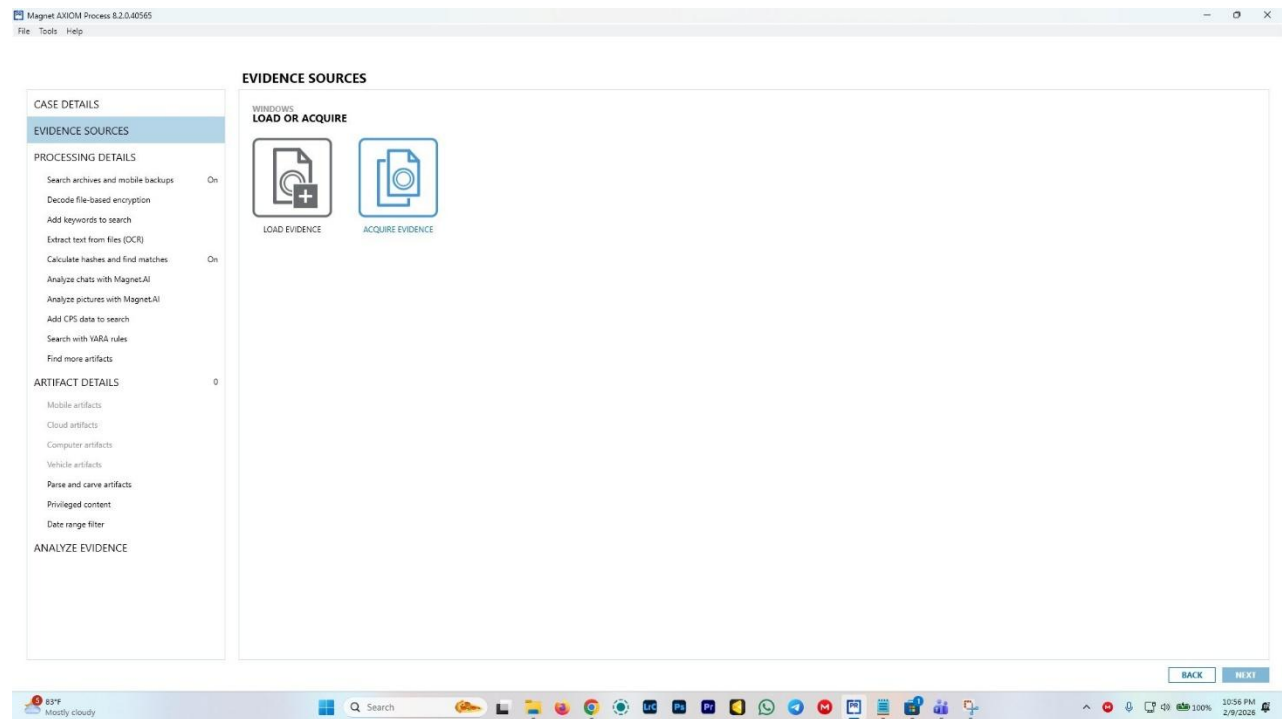
Windows was selected as the operating system from which the evidence source was acquired. This ensured the investigation remained strictly scoped to the target device and prevented the inclusion of unrelated storage media.

Evidence source selection



During the evidence source phase, Acquire Evidence was selected, as the data was to be acquired directly from the target device rather than loaded from a previously acquired source. The target device, a USB flash drive, was then selected from the list of mounted drives displayed in Magnet AXIOM Process.

The specific target, which was a flash drive of 8gb capacity, was identified in the listed mounted drives and selected for evidence acquisition.



Under Image Type, still within the evidence source phase, Full was selected to create an exact bit-by-bit copy of the flash drive. The E01 format was chosen for the acquisition, as it is a widely accepted forensic standard for disk imaging.

Processing options were configured to support the recovery of deleted and formatted data. Relevant artifact categories were enabled to allow examination of both allocated and unallocated space on the drive.

Finally, the evidence source was added to the case file.

Target flash drive selection

CASE DETAILS

EVIDENCE SOURCES

PROCESSING DETAILS

ARTIFACT DETAILS

ANALYZE EVIDENCE

COMPUTER

SELECT DEVICE

DRIVE

Name: F: Windows Basic Data (475.91 GB)

Type: 475.91 GB

Serial Number: 1BD9DC7B32A463AC66DB3C28A34FE31176FA100000

DRIVE

Name: PhysicalDrive3 Realtek RTL9210C NVME SCSI Disk Device (476.94 GB)

Type: External hard disk media

Serial Number: 476.94 GB

012345678962

DRIVE

Name: D: IFS (NTFS or HPFS) (476.94 GB)

Type: 476.94 GB

Serial Number: 61A2469031790B5EBB47FF1EC614C66773C000000

DRIVE

Name: PhysicalDrive1 NVMe SAMSUNG MZVLB512 SCSI Disk Device (476.94 GB)

Type: Fixed hard disk media

Serial Number: 476.94 GB

0025_3887_8183_3268

DRIVE

Name: E: FAT32 (LBA) (7.5 GB)

Type: 7.5 GB

Serial Number: 9C1D0BFC152AB4A857C2B955A64C96A110FF0000

DRIVE

Name: PhysicalDrive2 VendorCo ProductCode USB Device (7.5 GB)

Type: Removable Media

Serial Number: 5280031141739386675

83°F Mostly cloudy

Search

10:37 PM 2/9/2026

ANALYZE EVIDENCE

SOURCES TO PROCESS

Type	Image - location name	Evidence number	Search type	Start date/time - local time	End date/time - local time	Duration	Status
11	E: FAT32 (LBA) (7.5 GB)	7.5 GB Full Image	Full				Ready to image

EVIDENCE SOURCES

COMPUTER

SELECT IMAGE TYPE

Please select the type of images you want to acquire:

Full

☒ Entire contents of the drive in E01 format [More info](#)

☐ Entire contents of the drive in raw format [More info](#)

Files and Folders

☐ Acquire selected files and folders as in a container to search. [More info](#)

Search type

Select the search type for the source location:

Source location: E: FAT32 (LBA) (7.5 GB)

Search type: Full

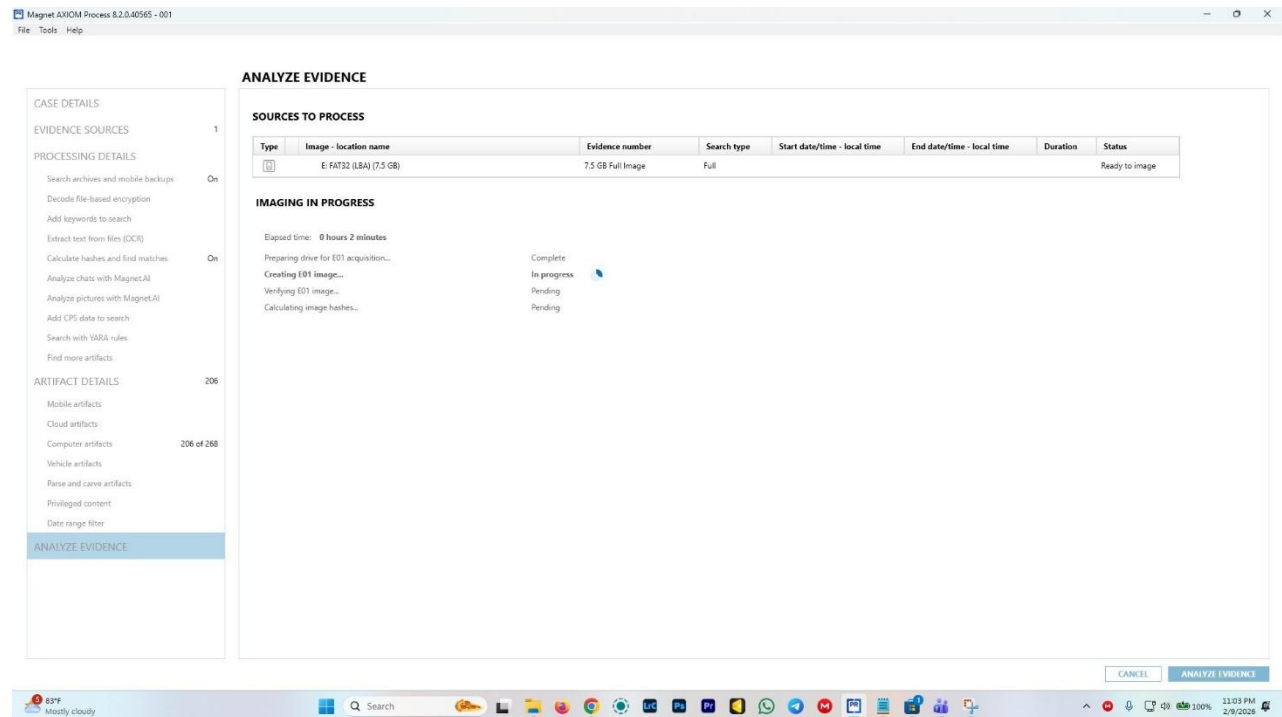
OKAY

4. Processing and Analysis

AXIOM Process analyzed the flash drive, examining the file system structure and residual data blocks. Despite the drive having been formatted, traces of previously stored data were detected during this stage.

AXIOM Process analyzing to recover artifacts. The bigger the volume of the target storage and the artifacts to be recovered. The longer the process is based on the specification of the host machine of AXIOM.

AXIOM processing results



5. Data Recovery and Examination Phase

The processed case was opened in Magnet AXIOM Examine automatically once AXIOM Process completed its task.

This phase focused on reviewing recovered artifacts in detail and validating their relevance to the investigation.

Recovered files and folders were identified within the processing results, which were automatically transferred from AXIOM Process to AXIOM Examine. The presence of reconstructed file entries indicated that the formatting process did not fully overwrite the underlying data.

AXIOM Examine categorized the results into the following sections:

- Case Overview, which provided details such as the case summary notes, the examiner's name, the case number, and the date and time of the acquisition process.
- Evidence Overview, which presented information about the target evidence source, including disk capacity and file system format.
- Place to Start, which highlighted key findings under Artifact Categories, showing the number of artifacts recovered, their types, and the associated operating system.

AXIOM Examine artifact categories and overview

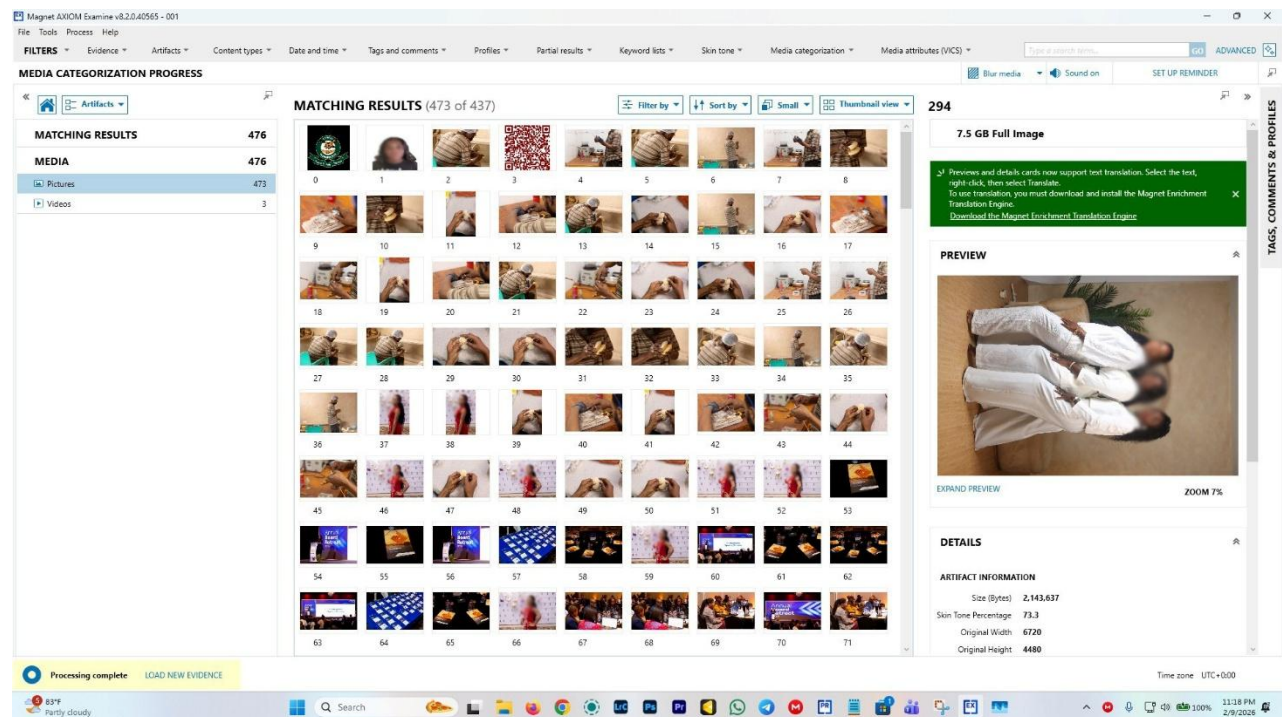
The screenshot displays the Magnet AXIOM Examine v8.2.0.40565 - 001 interface. The interface is divided into several sections:

- CASE OVERVIEW**: Includes a sidebar with 'EVIDENCE SOURCES' (1) and 'INSIGHTS' (0). The main area contains 'CASE SUMMARY NOTES', 'CASE PROCESSING DETAILS' (Case Number: 001, Scanned by: Julian Derry, Scan date/time: 2/9/2026 11:00:38.514 PM, Scan description: Recovering deleted files from a flashdrive), 'COLLABORATE USING MAGNET REVIEW', 'INVESTIGATE WITH MAGNET COPILOT', and 'CASE INFORMATION'.
- EVIDENCE OVERVIEW**: Shows '7.5 GB FULL IMAGE (149)' with a description, location (7.5 GB Full Image.001), platform (Computer), and process method (Parsing and carving). It includes a 'CHANGE PICTURE' button.
- PLACES TO START**: Includes 'ARTIFACT CATEGORIES' (Evidence source: All, Number of artifacts: 481, Media: 439, Custom: 23, Refined Results: 10, Web Related: 6, Documents: 2, Operating System: 1), 'TAGS AND COMMENTS', 'CPS DATA MATCHES', 'MAGNET.AI CATEGORIZATION', 'KEYWORD MATCHES', 'POTENTIAL CLOUD EVIDENCE LEADS' (No cloud evidence leads were found), 'MEDIA CATEGORIZATION', and 'PROFILES'.

The bottom status bar shows 'Processing complete', 'LOAD NEW EVIDENCE', 'Time zone: UTC+000', and '11:13 PM 2/9/2026'.

6. Artifact Verification

Recovered files were previewed and reviewed alongside available metadata. This step confirmed the integrity of the artifacts and verified that they originated from the formatted USB flash drive.



7. Findings and Conclusion

The examination confirmed that data could be successfully recovered from the formatted flash drive. This demonstrates that formatting alone is insufficient for permanent data removal and highlights the effectiveness of Magnet AXIOM in removable media forensics.

8. Final Assessment

This case reinforces a key forensic principle: data often remains recoverable after formatting. Proper data sanitization methods are required when permanent data destruction is necessary.

9. Tools Used

Magnet AXIOM Process & Magnet Examine